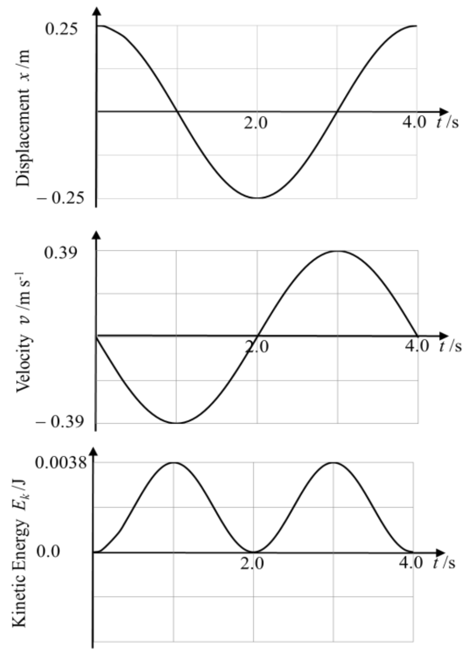
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)	$T = 2\pi\sqrt{\frac{l}{g}}$ $\omega = \frac{2\pi}{T} = \sqrt{\frac{g}{l}}$ combining formulae (1) $= \sqrt{\frac{9.81}{4.0}}$ = 1.57 rad s^{-1} (1) convincing		2		2	2	
		(ii)	$v_{max} = \omega A$ (1) v_{max} [= 1.57 × 0.25] = 0.39 m s^{-1} (1)		2		2	1	
		(iii)	$E_k = \frac{1}{2}mv^2$ **and** $v = (-)A\omega \sin \omega t$ / $v = (-)0.39 \sin 1.57t$ (1) $E_k = \frac{1}{2} \times 0.05 \times (0.39)^2 \sin^2 1.57t$ (1) $= 3.8 \times 10^{-3} \sin^2 1.57t$ [convincing]	1	1		2	1	
		(iv)	 Phase of the velocity graph (1) Period of the kinetic energy graph (1) Kinetic energy curve always ≥ 0 with shape and phase (1)						